The universe as a dynamical system: a problem of time for Brading's presentism

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Abstract

Brading's law-constitutive presentism locates the present of each thing in the spatiotemporal region needed to sustain it as a dynamical system, and she says that the principle of unity behind it applies "all the way up to the material universe as a whole." Her P1** requires every existing thing's present to be less than the whole history. This note asks what that requires of the universe itself. Within special relativity the universe's present is the whole history, a proper region whose cost Brading has not discussed, or absent; or the universe is outside the quantifier's domain. In general relativity the question becomes a version of the problem of time. For a spatially closed universe, the canonical options are: an internal time whose reduced Hamiltonian is time dependent, so that, on an energy reading, the universe fails Brading's conservation-based test of unity; an ideal clock that restores a conserved Hamiltonian but must be reference matter of a phantom kind, added for the purpose and, in the dust version, tunable to observational insignificance; cosmic time along the Hubble flow, which keeps standard cosmology but is fixed by an idealised model and a coordinative definition rather than by a sustaining system; or no time at all, in which case the universe has no present and P1** must be restricted. Brading's local view avoids the Lorentz-symmetry objection that Wüthrich raises against Monton's foliation presentism, but it meets the problem of time at the universe. The result is conditional and does not refute presentism. Its core is that on none of these routes does the dynamics of a sustaining system fix the universe's present: the decisive question is not only whether there are genuine subsystems, but what the universe's own present is.

1 The texts and the question

Katherine Brading's *Presentism as an Empirical Hypothesis* and *Physically Locating the Present* offer one of the most careful attempts to keep a presentist ontology alive in relativistic physics (Brading, 2013, 2015). She rejects the assumption that spacetime is the ontological principle of unity of what exists, and replaces it with a law-constitutive principle: the dynamical laws constitute genuine unities, and the present of a system is the region needed to sustain its dynamics:

> The present is a spatiotemporal region of whatever size is necessary to sustain the dynamical system in question (Brading, 2013, p. 1109).

The 2015 paper states the position as P1**:

> (P1**) For each and every thing that exists, that thing exists only presently, where for each thing the spatiotemporal extent of the present for that thing is dependent on the dynamics and is (a) less than the entire spatiotemporal evolution of the universe as represented in Minkowski spacetime, and (b) less than the entire spatiotemporal evolution of that thing as represented in Minkowski spacetime (Brading, 2015, p. 17).

She adds: "In order to avoid the block universe, all we need is (a)" (Brading, 2015, p. 17). And she places the decisive empirical question elsewhere, in the subsystems: whether there are genuine subsystems "smaller in spatiotemporal extent than the entirety of spacetime" is a matter of the details of the dynamics, and "only if there are such can the version of presentism offered here survive" (Brading, 2015, p. 18).

The scope of the principle is wider than the subsystems. Brading writes that the law-constitutive approach gives "a principle of ontological unity that applies across the board to simple things, composite systems, and component parts of those systems, and which applies all the way up to the material universe as a whole. Insofar as what exists is what is one, the law-constitutive approach to unity is a metaphysical thesis about what exists" (Brading, 2015, p. 16). This note asks what P1** requires of the universe itself, first within special relativity, the setting of Brading's texts, and then in general relativity, where the universe is standardly treated as a dynamical system in its own right.

The contribution is a bridge, not a new objection family. Savitt has already objected that every dynamical system has a sustaining region and that it is unclear when two systems share a present (Savitt, 2021, §3.3). Monton and Wüthrich have debated whether presentism can live in quantum gravity through a preferred foliation (Monton, 2006; Wüthrich, 2010). Isham, Thiemann, and Bamonti and Thébault have analysed time for the universe in canonical gravity and cosmology (Isham, 1991; Thiemann, 2006; Bamonti and Thébault, 2025). None of these discusses Brading's law-constitutive presentism or the universe's own present. The claim here is that, applied to the totality, Brading's own unity criterion meets the problem of time, and that the ways out each carry a definite cost.

2 The universe as a genuine unity

Brading's working premise is that "what exists (what is real) are the genuine unities," and that "what is 'present' depends on what is real (i.e. what exists, the genuine unities), and what is real depends on the dynamical laws" (Brading, 2015, p. 17). This note grants the premise. The question is what follows for the universe if it is granted.

Two further claims are needed for the universe to be an existing thing. The first is that the universe is a genuine unity. The scope claim above gives the principle's intended reach, not a verdict; a principle of unity can apply to a collection and find that it is not one. The textual support for the verdict is elsewhere: "The dynamics grounds the unity of what there is, both of the parts and of the whole (consisting of interacting parts)" (Brading, 2013, p. 1110); and the 2013 footnote says the law-constitutive approach covers "the unity of simple things, composite things, and the totality of existing things," while adding that "different principles of unity might be used for the different cases" (Brading, 2013, p. 1105, n. 3).

The second claim concerns the test. Brading associates unity with the conservation of total quantity of motion when a system is isolated from external interactions (Brading, 2013, p. 1105), and with a composite whole that conserves its total quantity and direction of motion "when that system as a whole is free from external interactions" (Brading, 2015, p. 16). The universe has no external interactions. In special relativity, a closed distribution of matter has a conserved total energy-momentum, so the universe passes the test. That is our application of her criterion, not her statement, and a reader may resist applying the isolation condition to the totality. If the premise is rejected, the universe falls outside the domain (§3, option 3).

For the general-relativistic discussion we need a regimentation of the test.

Definition 1 (Conservation test, regimented)**.** *A system passes the conservation test relative to a time parameter τ if its evolution in τ is generated by a quantity that is conserved along that evolution, that is, by a Hamiltonian with no explicit τ dependence.*

This is a bridge premise. It extends Brading's Newtonian and special-relativistic criterion to a setting she does not discuss, and it is the most contestable step in the argument. Brading's "quantity of motion" is Newtonian momentum, not energy. Definition 1 reads the test through the generator of time evolution, because that is the quantity whose conservation is at issue when a time is chosen. On a momentum reading the test does no work for a spatially closed universe: the momentum constraints of canonical general relativity make its total momentum vanish identically, so it passes trivially. Section 4 therefore separates the costs that depend on the energy reading from the one that does not, namely that the dynamics does not single out the universe's present.

3 The universe's present in special relativity

Grant that the universe is a genuine unity and an existing thing. P1** is universal: "for each and every thing that exists" (Brading, 2015, p. 17). Write $\mathcal{W}$ for the whole history in the common Minkowski representation. Four cases are exhaustive. Following Brading, "local" means a proper part of the block (Brading, 2015, p. 17); below, (2) is split by whether the proper region is spatially global.

(1) The present is $\mathcal{W}$. If the region needed to sustain the universe is its entire evolution, clause (a) is false of the universe, and P1** is false on its unrestricted reading. Brading herself says that if a system's needed region "is the entire spatiotemporal history of the universe," then "we are back to the block universe" (Brading, 2015, p. 17).

(2) The present is a proper region. *(2a) Spatially global.* If "needed to sustain" means "the data needed to determine the evolution," a Cauchy slice or a thin global slab would do. The cost is Brading's rejection of a global present: "the idea of a spatially global present common to all systems is not preserved in this version of presentism, but the rejection of a spatially global present is one of the central lessons of special relativity" (Brading, 2015, p. 17). The qualifier "common to all systems" leaves room for a global present for the universe alone. In Minkowski spacetime the dynamics can even select one: a closed system with timelike total energy-momentum has a rest frame. But the result is a spatially global present, in tension with the second clause of the passage, and Brading does not say how it relates to the local presents of the subsystems. *(2b) Local.* A patch might sustain the universe's unity if the laws determine the rest from it, but on Brading's account of aggregation "the members must stand in relations to one another, and those relations must be necessary and sufficient for the collection to form a unity" (Brading, 2015, p. 16). A local region must then encode the relations among all members; the text does not say how.

(3) The universe is outside the domain. Either (3a) it is not a genuine unity, and so, given the premise, does not exist, contrary to the scope claim; or (3b) "thing" is restricted to proper unities, and the quantifier no longer covers the whole of what exists. In both cases the view becomes a thesis about proper subsystems.

(4) The universe has no present. Brading's present is the region needed to sustain a *dynamical system*. If the universe is a unity but not a dynamical system in her sense, it has no sustaining

region, and P1** is false or undefined for it. Brading's own defeat condition speaks against this option being free: it describes the scenario in which "the only dynamical system that there is, is this entirety" (Brading, 2013, p. 1109), so it treats the totality as a dynamical system with a sustaining region.

Within special relativity, option (2a) is the natural home for a defender, and its cost is a global present for the universe. In special relativity the rest frame of the total energy-momentum supplies the slice. In general relativity there is no such frame for a closed universe, and the choice of slice becomes a specific physical problem.

4 The universe's present in general relativity

In canonical general relativity the Hamiltonian of a spatially closed universe is a sum of constraints and vanishes on solutions; there is no external time, and "nothing seems to happen" until a time is extracted from the variables themselves (Thiemann, 2006, §1). A spatially global present for the universe, option (2a), is then a region bounded by leaves of some global time function. Which function? Isham distinguishes two broad strategies: reduce the theory to true canonical form by choosing a time before quantising, or quantise all variables and impose the constraints, leading to the Wheeler–DeWitt equation (Isham, 1991, pp. 196–198). Combining these with the cosmological literature gives four routes for the universe's present. The question for each is whether the universe, so timed, passes the conservation test.

(a) An internal time. Choose a time variable from the gravitational or matter degrees of freedom and solve the constraint for its conjugate momentum, which becomes the reduced Hamiltonian. In a closed Friedmann–Lemaître–Robertson–Walker minisuperspace Isham does this for the radius of the universe, for its conjugate momentum, and for a scalar field, and lists among the general problems that "the reduced Hamiltonian is time dependent" (Isham, 1991, pp. 199–200). He comments:

> A time-dependent Hamiltonian means that energy can be fed into, or taken out of, the quantum system. In a normal physical situation this arises because the system is not closed, and then the time-dependence of the Hamiltonian is determined by the environment. However, a compact three-manifold Σ (the "universe") has no external environment, and so the time-dependence is perhaps a little odd (Isham, 1991, p. 200).

Wüthrich reports the same point against the constant-mean-curvature route to a presentist foliation (Wüthrich, 2010, p. 274). For Brading the oddity is more than odd. A universe whose evolution in its own global time is generated by a time-dependent Hamiltonian fails the conservation test of Definition 1. It has a present, but by her criterion it is not the kind of isolated whole that conserves its total quantity of motion. Isham adds a second cost: "There is no natural choice for the time variable and no reason to suppose that different choices will give equivalent quantum theories" (Isham, 1991, p. 200).

(b) An ideal clock. The time dependence is not universal. In Isham's own scalar-field example the reduced Hamiltonian, $H_\phi = \pm(a^2\pi_a^2/12 + 12a^4 - 2a^6V(t))^{1/2}$ (Isham, 1991, eq. 5.2.8), has no explicit time dependence when the potential V vanishes. Thiemann generalises the point: if the Lagrangian of a scalar field coupled to general relativity contains only derivatives, the theory deparametrises completely, and there is "a distinguished notion of time generated by an invariant,

positive and time independent physical Hamiltonian" (Thiemann, 2006, p. 20). On this route the universe has a global time and passes the conservation test.

The cost is physical. To keep the physical Hamiltonian close to that of standard cosmology the clock Lagrangian must be of Dirac–Born–Infeld type, which for small gradients is a massless field "with the wrong sign, i.e. it is a phantom," and "non-trivial potentials are forbidden" (Thiemann, 2006, p. 5). The phantom's negative energy must be compensated by other matter, and the concrete model implies that the universe recollapses (Thiemann, 2006, pp. 9–10). Above all, Thiemann claims that "whatever matter is used for deparametrisation, there will be corrections to the standard FRW equations. This should have observable consequences!" (Thiemann, 2006, p. 9).

That general claim is too strong as an empirical cost. Using pressureless dust as the clock, in the manner of Brown and Kuchař (Brown and Kuchař, 1995), Giesel, Hofmann, Thiemann and Winkler obtain a physical Hamiltonian that "is simply the spatial integral of the energy density," and whose evolution equations match Einstein's "up to modifications proportional to the energy momentum density" of the dust, so that "the influence of the dust can be tuned away arbitrarily" (Giesel et al., 2007a, p. 4). For a spatially flat Friedmann–Lemaître–Robertson–Walker background the resulting equations "formally agree with the standard FRW equations" (Giesel et al., 2007b, p. 11), and the corrections to linear perturbations "decay in the late universe at least as $1/a$" (Giesel et al., 2007b, p. 4). What survives is the character of the clock. The dust "must be a 'phantom dust'," because with ordinary dust "the physical Hamiltonian would come out negative definite rather than positive definite. Or equivalently, physical time would run backwards rather than forward"; its energy density can be arbitrarily small but "must not vanish" (Giesel et al., 2007a, p. 4). On route (b), then, the universe has a global present and passes the conservation test, but the present is fixed by a reference medium of negative energy, posited to serve as clocks and rods, whose observable trace can be made as small as one likes.

None of this depends on spatial flatness. The framework of Giesel and co-authors is set up for a three-manifold "of arbitrary topology," and the case without boundary, which includes the spatially closed universe, is the one they treat by default: "As long as the dust space ... has no boundary, H is functionally differentiable, which we always assumed so far" (Giesel et al., 2007a, pp. 12, 36). Thiemann's phantom scalar likewise works for compact σ without boundary (Thiemann, 2006, p. 26). Only their cosmological comparisons are made for flat backgrounds. In Isham's closed model the point is elementary. Add homogeneous comoving dust with proper time T and conjugate momentum P_T. For homogeneous dust the Brown–Kuchař energy density reduces to $|P_T|$, so the constraint (5.2.5) becomes $C(a, \pi_a, \phi, \pi_\phi) \pm P_T = 0$, where C is Isham's left-hand side, curvature term included. The physical Hamiltonian $\mp P_T = C$ contains no T, for $k = 1$ exactly as for $k = 0$, and the sign is the one Giesel and co-authors show forces phantom dust. This extension to the closed case is ours, not stated in the sources. The empirical exposure is real for Thiemann's phantom scalar and arbitrarily weak for phantom dust. The cost that does not weaken is that the universe's present is fixed by an added reference system, not by the region needed to sustain the universe's own dynamics.

(c) Cosmic time. Bamonti and Thébault defend a conservative alternative: read the time in the Friedmann equations as proper time along the Hubble flow, in synchronous gauge, so that the equations describe the correlation between two partial observables, the Hubble parameter and cosmic time (Bamonti and Thébault, 2025, §§4–5). The cost is that "there is no physical system which even approximately measures proper time along the Hubble flow" (Bamonti and Thébault, 2025, p. 5). Cosmic time is recovered as a model-based global average, with the clock hypothesis acting as a coordinative definition in Reichenbach's sense (Bamonti and Thébault,

2025, pp. 19–20). They do not address whether the universe passes a conservation test in this gauge, and nothing here is a criticism of their proposal. On the energy reading the answer depends on implementation. Taken as a gauge fixing, as they take it, the evolution in cosmic time is generated by the Hamiltonian constraint itself, which vanishes, so the test has no non-trivial content. Implemented instead by deparametrising with the comoving matter, whose proper time is cosmic time, route (c) becomes an instance of route (b) with dust as the clock. The actual cosmological matter cannot play that role directly: it has positive energy, and ordinary dust would make physical time run backwards (Giesel et al., 2007a, p. 4). The point is narrower. If the universe's present is a slice of constant cosmic time, it is fixed by an idealised homogeneous model and a coordinative definition. It is not "the region necessary to sustain the dynamical system" in Brading's sense, since no actual system's dynamics singles it out.

(d) No time. If no time is extracted, the constraints are imposed on the states and the universe is described by a solution of the Wheeler–DeWitt equation with no evolution parameter (Isham, 1991, §5.3). The universe may then count as a conserved whole, but it has no present. This is option (4) of §3, now physically motivated, and P1** must be restricted to subsystems with internal clocks.

Route	Global present	Conservation test	Cost for Brading
(a) internal time	yes	fails (energy reading)	universe is not a unity
(b) ideal clock	yes	passes	present fixed by added phantom reference matter
(c) cosmic time	yes	trivial or as (b)	present fixed by model and convention
(d) no time	no	passes trivially	P1** restricted

The argument has the following structure. Grant that the universe is a genuine unity and a thing (§2). Then, for a spatially closed universe in canonical general relativity, either the universe has no present (d), or its present is a global slice fixed by a choice of time. On any reading of the unity test, that choice is not made by the dynamics of a sustaining system: there is "no natural choice for the time variable" (a), the clock must be special matter added for the purpose (b), or the slice is fixed by an idealised model and a coordinative definition (c). On the energy reading of Definition 1 there are further costs: route (a) loses the universe's unity, and route (b) keeps it only by adding a phantom reference medium. The note does not claim the four routes are exhaustive of all possible physics, only that they are the routes in the cited literature and that the text gives the defender none of them for free.

5 Monton, Wüthrich and the local present

The debate over presentism in quantum gravity has concerned a global present. Monton grants that presentism is incompatible with special and general relativity, and argues that it is compatible with quantum theories of gravity that use a fixed foliation, such as constant-mean-curvature canonical gravity, on a theoretical hypothesis that one leaf represents the existing events (Monton, 2006). Wüthrich replies that fixed-foliation theories break a Lorentz symmetry that is "fantastically well confirmed," and that the constant-mean-curvature approach is limited in scope and does not by itself supply a present that underwrites becoming (Wüthrich, 2010, pp. 269–275).

Brading's view is designed to avoid that exchange. It does not posit a foliation; presents are local, dynamically individuated and not common to all systems. Wüthrich's classification of presentist responses to special relativity covers frame-relative existence, the point present and the past-light-cone present (Wüthrich, 2010, pp. 263–264); it does not cover a present fixed by a system's dynamics. So the Lorentz-symmetry objection does not apply to Brading's subsystems.

It applies, in effect, to the universe. Routes (a) to (c) of §4 all give the universe a global slice, and route (b) is precisely a fixed-foliation theory of the kind Monton relies on. Brading's local presentism therefore inherits the Monton–Wüthrich problem at exactly one scale, the scale at which her principle of unity is said to apply "all the way up."

6 The defeat condition, relocated

Brading's 2013 defeat condition describes one scenario:

> It might turn out that, for whatever system we try to consider, the size of the spatiotemporal region necessary to sustain it is the entire history of the universe, encompassing all that ever has been and ever will be. In other words, the only dynamical system that there is, is this entirety, and there are no genuine subsystems of the universe. If that turns out to be the case, then this version of presentism is defeated and the block universe triumphs (Brading, 2013, p. 1109).

The universe case is independent of it. Even if there are genuine subsystems with proper presents, the universe remains a candidate unity with a present to fix. Brading's empirical question, whether there are genuine subsystems, does not decide what that present is. Section 4 shows that the universe case has a different character. Its empirical exposure is limited: Thiemann's phantom scalar alters late-time cosmology, but a phantom-dust clock can be tuned to observational insignificance, and cosmic time keeps standard cosmology. What the universe case exposes is rather a gap in the law-constitutive account itself. On routes (b) and (c) the universe's present is fixed by an added reference medium or by an idealised model; on routes (a) and (d) the universe loses its unity or its present. Brading calls her view an empirical hypothesis. Applied to the totality, its burden is less a prediction than an account of what fixes the present of the whole.

7 The retreat to subsystems and the coverage model

The natural reply is to restrict P1** to proper subsystems (option 3b, or route (d)). The restricted view no longer says that every existing thing exists only presently. A defender who instead keeps the global claim and tries to secure it through clauses (a) and (b) for the subsystems faces a further, independent problem: the clauses do not imply that any event lies outside every present.

Proposition 2 ((a) and (b) do not imply a proper union)**.** *There is a model in which (a) and (b) hold for every system and* $\bigcup_i \mathcal{U}_i = \mathcal{W}$.

Proof. Let $\mathcal{W} = \{a, b, c, d\}$. Let system 1 have history $\mathcal{H}_1 = \{a, b, c\}$ and present $\mathcal{U}_1 = \{a, b\}$, and system 2 have history $\mathcal{H}_2 = \{b, c, d\}$ and present $\mathcal{U}_2 = \{c, d\}$. Then $\mathcal{U}_i \subsetneq \mathcal{W}$ and $\mathcal{U}_i \subsetneq \mathcal{H}_i$ for $i = 1, 2$, but $\mathcal{U}_1 \cup \mathcal{U}_2 = \mathcal{W}$. □

The model is set-theoretic, not dynamical; it regiments "less than" as proper inclusion; and it uses a common event domain $\mathcal{W}$. Brading's 2013 note 7 denies determinate ontological spatial or

temporal relations between systems: "All the ontology is carried by the dynamics, and we frame the dynamics spatiotemporally, but ontologically, the dynamics requires no such spatiotemporal underpinning" (Brading, 2013, p. 1109, n. 7). The model therefore treats $\mathcal{W}$ as the common epistemic representation in which the comparison is stated. With those assumptions, per-system properness is coverage-neutral, a formal counterpart of Savitt's observation that every dynamical system has a sustaining region (Savitt, 2021, §3.3).

8 Objections and limits

Brading's view is special-relativistic. Her texts work in Minkowski spacetime, and P1** is stated there. Section 3 is internal to that setting. Section 4 is an extension, and its bridge premise, Definition 1, is ours. A defender may reject it, for instance by holding that the universe's unity does not require a conserved generator of its evolution. That is a substantive revision of the criterion Brading takes from Newton, and the note's claim is that some such revision, or one of the costs in the table, is required.

Momentum, not energy. If the unity test is read as momentum conservation, as Brading's Newtonian source suggests, the universe passes it trivially, and the failure recorded for route (a) in the table disappears. The selection cost remains: on no route examined does the dynamics of a sustaining system fix the universe's present. The argument is weaker on this reading but does not lapse.

The routes may not be exhaustive. The note claims only that the routes in the cited literature each carry a cost, not that no other route exists.

Scope of the physics. Isham's examples are closed minisuperspace models. The dust and phantom-scalar constructions hold for closed spatial topology, but their cosmological comparisons use spatially flat backgrounds, and our closed-model extension (§4) covers only the homogeneous background, not perturbations; and the dust analysis shows that Thiemann's claim of observable corrections for every clock does not hold in an empirically significant form. Bamonti and Thébault do not discuss conservation in synchronous gauge. Each of these is a point at which the argument could be weakened by further physics.

Universe and representation. Brading's note 7 suggests that the spatiotemporal representation carries no ontology of its own. If so, the global slice of routes (a) to (c) is a representational device, and the universe's present is not a region at all. Then route (d) or option (4) holds, and the restriction of P1** follows.

Priority. No published statement of this argument was found in the sources read: Brading 2013 and 2015, Savitt 2021, Monton 2006 (2005 preprint), Wüthrich 2010, Isham 1991 (pp. 192–202), Thiemann 2006 and Bamonti and Thébault 2025, all at full-text depth; Balaguer's *How to Be a Presentist* (Balaguer, 2026) and Mason's empirical presentism (Mason, 2026) only at abstract depth, as is Brown and Kuchař (Brown and Kuchař, 1995); Giesel, Hofmann, Thiemann and Winkler (Giesel et al., 2007a,b) at the introductions and passages cited. Balaguer's chapter on special relativity is the main unread risk. A prior statement would defeat the priority claim, not the argument.

What the note does not show. It does not show that presentism is false, that eternalism is true, or that Brading's subsystem presents are incoherent. It does not show that the universe's present is $\mathcal{W}$, and it takes no position on the cosmology of routes (b) and (c).

9 Conclusion

Brading's principle of unity is offered for the whole of what exists, up to the material universe. Taken at that scale, P1** needs an account of the universe's own present. In special relativity the options are the whole history, a global slice of the kind she elsewhere rejects, an unexplained local region, exclusion from the domain, or no present at all. In general relativity the global-slice option becomes the problem of time. The universe can have a present and fail the conservation test (internal time), have a present and pass it by adding a phantom reference medium whose observable trace can be made arbitrarily small (ideal clock), have a present fixed by an idealised model and a coordinative definition (cosmic time), or have no present and leave P1** restricted to subsystems. Brading's local presentism escapes the Lorentz-symmetry objection to foliation presentism for every system except the universe. There, on every route examined, the present is fixed by something other than the region needed to sustain the universe's own dynamics, or there is no present at all. Her hypothesis remains empirical about subsystems; about the whole it owes an account that the law-constitutive principle does not yet give.

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